PEDIATRIC MEDICINE · NEPHROLOGY · LECTURE 1 · QUESTION BANK

Nephrology

محاضرة أمراض الكلى عند الأطفال — بنك الأسئلة

10	73	58
Past exam questions	NCLEX-style	Flashcards

المصادر اللي اتجمعت منها أسئلة المحاضرة دي:

أسئلة سابقة

جنب كل سؤال مصدره. لو السؤال اتكرر في كذا امتحان مكتوب كل مرة بمصدرها، وعليه علامة "تكرر". الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1 أسئلة سابقة

Nephrotic syndrome is primarily characterized by:

a. Hematuria
b. High creatinine levels
c. Massive proteinuria
d. Urinary tract obstruction

2 أسئلة سابقة

The most common cause of nephrotic syndrome in children is:

a. FSGS
b. Membranous nephropathy
c. Minimal change disease
d. Lupus nephritis

3 أسئلة سابقة

A key difference between nephritis and nephrosis is:

a. Foamy urine in nephritis
b. Inflammation in nephrosis
c. Edema in nephritis only
d. Inflammation is a main feature of nephritis

4 أسئلة سابقة

Which of the following is a complication of nephrotic syndrome?

a. Hypotension
b. Weak immune system (infection)
c. Hyperkalemia
d. Anemia

5 أسئلة سابقة

Which diagnostic tool confirms kidney tissue damage?

a. CT scan
b. Urinalysis
c. Blood test
d. Kidney biopsy

6 أسئلة سابقة

Nephritis is commonly caused by:

a. Hyperlipidemia
b. Autoimmune diseases
c. Minimal change disease
d. Nephrotoxic drugs only

7 Which medication is used to manage high cholesterol in nephrosis? أسئلة سابقة

a. Furosemide
b. Atorvastatin
c. Penicillin
d. Prednisone

8 In nephritis, which symptom is common? أسئلة سابقة

a. Massive proteinuria
b. Foamy urine
c. Hypertension
d. Severe hyperlipidemia

9 Which dietary recommendation is common for both nephritis and nephrosis? أسئلة سابقة

a. High-sodium intake
b. High-protein diet
c. Low-protein, low-sodium diet
d. No dietary restrictions

10 What is the purpose of corticosteroids in nephritis treatment? أسئلة سابقة

a. Lower blood pressure
b. Control fluid retention
c. Treat infections
d. Reduce inflammation

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

N1 Which function is NOT performed by the kidney? NCLEX

a. Activation of vitamin D (1-alpha hydroxylation)
b. Production of erythropoietin
c. Production of bile
d. Regulation of acid-base balance

N2 The structural and functional unit of the kidney is the: NCLEX

a. Glomerulus only
b. Nephron
c. Renal pelvis
d. Collecting duct

N3 The glomerular filtration barrier consists of (select all): SELECT ALL

a. Fenestrated endothelium
b. Glomerular basement membrane (negatively charged)
c. Podocytes with slit diaphragms
d. Renal capsule
e. Collecting duct epithelium

N4 Normal protein excretion in a child is: NCLEX

a. ≤ 4 mg/m²/hr (P/C ratio < 0.2)
b. > 40 mg/m²/hr
c. P/C ratio > 2
d. 3+ on dipstick

N5 The first-line imaging study for a child with a renal problem is: NCLEX

a. CT abdomen
b. Renal ultrasound
c. MRI
d. Renal biopsy

N6 **Hematuria of glomerular origin is suggested by (select all):** SELECT ALL

a. Cola / smoky urine
b. RBC casts
c. Dysmorphic RBCs
d. Clots in urine
e. Associated proteinuria

N7 **Renal biopsy is examined by (select all):** SELECT ALL

a. Light microscopy
b. Immunofluorescence
c. Electron microscopy
d. Culture
e. Gram stain

N8 **Which hormone produced by the kidney stimulates red blood cell production?** NCLEX

a. Renin
b. Erythropoietin
c. Aldosterone
d. Calcitriol

N9 **A child's urinalysis shows SG 1.010–1.025, pH 6, no protein, 2 RBCs/HPF. The nurse interprets this as:** NCLEX

a. Normal urinalysis
b. Glomerular hematuria
c. Nephrotic-range proteinuria
d. Urinary tract infection

N10 **Which test is used to detect vesicoureteral reflux?** NCLEX

a. Renal ultrasound
b. Voiding cystourethrogram (VCUG)
c. Plain X-ray KUB
d. Urine culture

N11 **Before a percutaneous renal biopsy, the nurse should check (select all):** SELECT ALL

a. Coagulation profile and platelets
b. Blood pressure
c. Signed consent
d. Serum lipase
e. Hemoglobin / blood grouping

N12 **After a renal biopsy, which finding requires immediate reporting?** NCLEX

a. Mild pink urine in the first void
b. Falling blood pressure with rising pulse
c. Mild discomfort at the site
d. Child asking for food

N13 **Bright red urine with clots in a child suggests bleeding from:** NCLEX

a. The glomerulus
b. The lower urinary tract
c. The tubules
d. The liver

N14 **Red urine with a negative dipstick for blood and no RBCs may be caused by (select all):** SELECT ALL

a. Beetroot ingestion
b. Rifampicin
c. Urate crystals in infants
d. Post-streptococcal GN
e. Food dyes

N15 **A 4-year-old with minimal change nephrotic syndrome has generalized edema. Which laboratory finding confirms the diagnosis?** NCLEX

a. Serum albumin 1.8 g/dL with heavy proteinuria
b. Low C3 complement level
c. RBC casts in urine
d. Serum potassium 6.2 mEq/L

N16 **Which findings are part of the definition of nephrotic syndrome? Select all that apply.** SELECT ALL

a. Hypoalbuminemia < 2.5 g/dL
b. Cholesterol > 200 mg/dL
c. Generalized pitting edema
d. Gross hematuria
e. Heavy (nephrotic-range) proteinuria

N17 **The physician orders prednisone 2 mg/kg/day for a child with nephrotic syndrome. The best indicator that the drug is effective is:** NCLEX

a. Increased appetite
b. Urine protein becoming negative / trace for 3 days
c. Rising blood pressure
d. Weight gain

N18 **A child with nephrotic syndrome suddenly develops fever and abdominal pain with tenderness. The nurse suspects:** NCLEX

a. Hypovolemic shock
b. Spontaneous bacterial peritonitis
c. Renal vein thrombosis
d. Steroid psychosis

N19 **Nursing priorities for an edematous child with nephrotic syndrome include (select all):** SELECT ALL

a. Daily weight at the same time
b. Measure abdominal girth
c. Turn frequently and protect edematous skin
d. High-salt diet to improve appetite
e. Keep away from children with infections

N20 **Which complication of nephrotic syndrome results from urinary loss of antithrombin III?** NCLEX

a. Infection
b. Thrombosis (DVT, renal vein)
c. Hypocalcemia
d. Hypertension

N21 **Renal biopsy before steroid treatment in nephrotic syndrome is indicated when (select all):** SELECT ALL

a. Onset below 1 year or above 10 years
b. Persistent hypertension
c. Low C3
d. Gross hematuria
e. Selective proteinuria in a 4-year-old

N22 **A child on long-term steroids for nephrotic syndrome is going home. Which instruction is most important?** NCLEX

a. Stop prednisone immediately if face becomes puffy
b. Test urine for protein daily and avoid live vaccines
c. Increase salt intake
d. Restrict all protein

N23 **Congenital nephrotic syndrome is defined as nephrotic syndrome that presents:** NCLEX

a. In the first 3 months of life
b. Between 1 and 10 years
c. After puberty
d. Only after a strep infection

N24 **Steroid-resistant nephrotic syndrome is failure to achieve remission after:** NCLEX

a. 1 week of prednisone
b. 4–6 weeks of daily prednisone
c. 6 months of therapy
d. One dose of methylprednisolone

N25 **Frequent relapses in nephrotic syndrome means:** NCLEX

a. ≥ 2 relapses within 6 months or ≥ 4 in 12 months
b. 1 relapse per year
c. Relapse during the first week of treatment
d. Persistent hematuria

N26 **The edema of nephrotic syndrome is mainly due to:** NCLEX

a. Increased capillary hydrostatic pressure
b. Decreased plasma oncotic pressure from hypoalbuminemia
c. Lymphatic obstruction
d. Heart failure

N27 **Where is edema of nephrotic syndrome first noticed in a child?** NCLEX

a. Feet in the evening only
b. Periorbital in the morning
c. Hands
d. Sacrum

N28 **Hyperlipidemia in nephrotic syndrome occurs because:** NCLEX

a. Fat intake is high
b. The liver increases lipoprotein synthesis in response to low albumin
c. Steroids are given
d. The kidney produces cholesterol

N29 **Side effects of long-term corticosteroid therapy that the nurse should monitor (select all):** SELECT ALL

a. Hypertension
b. Hyperglycemia
c. Cushingoid face and weight gain
d. Growth retardation
e. Hypoglycemia

N30 **A child with severe edema and low albumin is ordered IV albumin 20% followed by furosemide. The nurse's priority during infusion is:** NCLEX

a. Check for hypokalemia only
b. Monitor for fluid overload: BP, respiration, crackles
c. Encourage oral fluids
d. Restrict all activity

N31 **Which immunization is especially recommended for children with nephrotic syndrome when off high-dose steroids?** NCLEX

a. Pneumococcal vaccine
b. BCG only
c. Rabies vaccine
d. Yellow fever vaccine

N32 **Which nursing diagnosis has the highest priority for a child with anasarca due to nephrotic syndrome?** NCLEX

a. Excess fluid volume
b. Knowledge deficit
c. Disturbed body image
d. Activity intolerance

N33 **Which history is most expected in a child admitted with acute post-streptococcal glomerulonephritis?** NCLEX

a. Sore throat 1–2 weeks ago
b. Measles 6 months ago
c. Recent dog bite
d. Allergy to milk

N34 **Classic features of APSGN include (select all):** SELECT ALL

a. Cola / tea-colored urine
b. Periorbital edema
c. Hypertension
d. Massive selective proteinuria
e. Oliguria

N35 **A child with AGN complains of headache and blurred vision; BP 160/110. The nurse's priority is:** NCLEX

a. Encourage fluids
b. Notify the physician — suspect hypertensive encephalopathy
c. Give a high-protein snack
d. Ambulate the child

N36 **Which laboratory result supports the diagnosis of APSGN?** NCLEX

a. Low C3 with raised ASO titre
b. Serum albumin 1.5 g/dL
c. Cholesterol 450 mg/dL
d. Negative urine for RBCs

N37 **Fluid allowance for an oliguric child with AGN is calculated as:** NCLEX

a. Maintenance fluids plus 20%
b. Previous day's urine output + insensible loss (≈400 mL/m^2/day)
c. Unlimited oral fluids
d. Nothing by mouth for 48 hours

N38 **The earliest sign of improvement in a child with AGN is:** NCLEX

a. Disappearance of microscopic hematuria
b. Increased urine output (diuresis)
c. Normal C3
d. Weight gain

N39 **Nursing care of a child with AGN includes (select all):** SELECT ALL

a. Daily weight and strict I&O
b. BP monitoring every 4 hours
c. Low-salt diet
d. Bed rest during the acute oliguric phase
e. Encourage high-potassium foods

N40 **Indications for renal biopsy in suspected APSGN include:** NCLEX

a. C3 still low after 8 weeks or normal at onset
b. Tea-colored urine in the first week
c. Periorbital edema
d. Positive throat culture

N41 **The pathogenesis of APSGN is best described as:** NCLEX

a. Direct bacterial invasion of the kidney
b. Immune complex deposition in the glomeruli
c. Obstruction of the ureter
d. Toxic drug injury

N42 **Hypertension in acute glomerulonephritis is mainly due to:** NCLEX

a. Renin excess only
b. Salt and water retention from decreased GFR
c. Anxiety
d. Steroid therapy

N43 **Which complications of APSGN require ICU admission? (select all)** SELECT ALL

a. Hypertensive encephalopathy
b. Acute heart failure
c. Acute kidney injury with hyperkalemia
d. Mild periorbital edema
e. Microscopic hematuria

N44 **Antibiotic therapy in APSGN is given to:** NCLEX

a. Cure the kidney inflammation
b. Eradicate remaining streptococci and prevent spread
c. Lower blood pressure
d. Reduce proteinuria

N45 Which statement about the prognosis of APSGN in children is correct? NCLEX

a. Most children (> 95%) recover completely
b. Most progress to chronic kidney disease
c. Relapses are common
d. Steroids are needed for recovery

N46 A child with AGN is oliguric with K^+ 6.5 mEq/L. Which food should be avoided? NCLEX

a. Rice
b. Bananas and oranges
c. Apples
d. White bread

N47 Compared with nephrotic syndrome, acute nephritis has (select all): SELECT ALL

a. Gross hematuria
b. Hypertension
c. Low C3
d. Massive proteinuria
e. Marked hyperlipidemia

N48 The best way to evaluate fluid status in a child with AGN is: NCLEX

a. Daily weight on the same scale
b. Skin color
c. Thirst
d. Number of meals

N49 The most common organism causing UTI in children is: NCLEX

a. Klebsiella
b. Escherichia coli
c. Proteus
d. Pseudomonas

N50 Which presentation of UTI is typical in a neonate? NCLEX

a. Dysuria and frequency
b. Fever or temperature instability, poor feeding, jaundice
c. Flank pain
d. Enuresis

N51 Predisposing factors for UTI in children include (select all): SELECT ALL

a. Vesicoureteral reflux
b. Posterior urethral valves
c. Uncircumcised male infant
d. Constipation
e. Drinking plenty of water

N52 A urine culture from a suprapubic aspirate is diagnostic of UTI when it shows: NCLEX

a. > 100,000 CFU only
b. Any bacterial growth
c. Mixed growth of 3 organisms
d. No growth

N53 Which instruction helps prevent recurrent UTI in a school-age girl? NCLEX

a. Wipe from back to front
b. Void regularly and completely; drink enough fluids
c. Take bubble baths
d. Hold urine as long as possible

N54 **Signs pointing to upper UTI (pyelonephritis) rather than cystitis are (select all):** SELECT ALL

a. High fever with rigors
b. Loin pain / tenderness
c. Low-grade or no fever
d. Vomiting and toxic look
e. Suprapubic pain only

N55 **Which urine collection method has the highest contamination rate?** NCLEX

a. Suprapubic aspiration
b. Urinary catheter
c. Urine bag
d. Mid-stream clean catch

N56 **Vesicoureteral reflux increases the risk of:** NCLEX

a. Recurrent pyelonephritis and renal scarring
b. Nephrotic syndrome
c. Hypospadias
d. Wilms tumor

N57 **Indications for IV antibiotics in a child with UTI include (select all):** SELECT ALL

a. Age below 3 months
b. Toxic appearance or vomiting
c. Suspected pyelonephritis with dehydration
d. Mild dysuria in a 10-year-old
e. Urosepsis

N58 **The nurse teaches parents to collect a clean-catch urine specimen. Which instruction is correct?** NCLEX

a. Collect the first drops of urine
b. Clean the genital area, discard the first part, then collect midstream
c. Keep the sample at room temperature for 6 hours
d. Use any clean jar at home

N59 **Which finding on urinalysis supports UTI?** NCLEX

a. Positive leukocyte esterase and nitrites
b. Heavy proteinuria
c. RBC casts
d. High specific gravity only

N60 **Prophylactic antibiotics for recurrent UTI are usually:** NCLEX

a. Low-dose nitrofurantoin or TMP-SMX at night
b. IV ceftriaxone daily
c. High-dose amoxicillin three times daily
d. Gentamicin IM weekly

N61 **A child with severe dehydration from gastroenteritis becomes oliguric. This type of AKI is:** NCLEX

a. Pre-renal
b. Renal (intrinsic)
c. Post-renal
d. Chronic

N62 A child with AKI has serum K^+ 7.2 mEq/L with ECG changes. The first drug to give is: NCLEX

a. Furosemide
b. Calcium gluconate 10% slowly IV
c. Sodium bicarbonate orally
d. Kayexalate rectally

N63 Indications for dialysis in AKI include (select all): SELECT ALL

a. Fluid overload with pulmonary edema
b. Refractory hyperkalemia
c. Uremic encephalopathy
d. Severe metabolic acidosis not responding
e. Mild rise in urea only

N64 Which nursing measure is most important for a child in the oliguric phase of AKI? NCLEX

a. Encourage oral fluids freely
b. Strict intake and output with daily weight
c. High-potassium diet
d. Ambulate frequently

N65 Laboratory findings expected in AKI include (select all): SELECT ALL

a. Raised urea and creatinine
b. Metabolic acidosis
c. Hyperkalemia
d. Hypophosphatemia
e. Hypocalcemia

N66 A child with posterior urethral valves develops anuria. This AKI is: NCLEX

a. Pre-renal
b. Renal
c. Post-renal (obstructive)
d. Hepatorenal

N67 Oliguria in a child is defined as urine output less than: NCLEX

a. 0.5–1 mL/kg/hr
b. 3 mL/kg/hr
c. 2 L/day
d. 5 mL/kg/hr

N68 Which drugs are nephrotoxic and can cause intrinsic AKI? (select all) SELECT ALL

a. Aminoglycosides (gentamicin)
b. NSAIDs
c. Contrast media
d. Paracetamol in normal dose
e. Amphotericin B

N69 Hemolytic uremic syndrome causes AKI with: NCLEX

a. Hemolytic anemia, thrombocytopenia and renal failure after bloody diarrhea
b. Massive proteinuria and edema
c. Fever and flank pain
d. Hydronephrosis

N70 ECG change typical of hyperkalemia is: NCLEX

a. Peaked (tall tented) T waves
b. Prominent U waves
c. ST depression only
d. Prolonged QT from hypocalcemia only

N71 During the diuretic (recovery) phase of AKI, the nurse should watch for: NCLEX

a. Hyperkalemia
b. Dehydration and hypokalemia
c. Fluid overload
d. Hypertension from salt

N72 Peritoneal dialysis is often preferred in infants because: NCLEX

a. It is technically simple and needs no vascular access
b. It removes fluid faster than any method
c. It has no infection risk
d. It does not need a catheter

N73 Diet for a child with AKI in the oliguric phase should be (select all): SELECT ALL

a. Adequate calories mainly from carbohydrate and fat
b. Restricted potassium
c. Restricted phosphate
d. High sodium
e. Controlled protein

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

Kidney functions	Water & electrolytes, acid-base, waste excretion, BP (renin), erythropoietin, vitamin D activation.	**Nephron**	Structural & functional unit; ~1 million per kidney; glomerulus + tubules.
Cortex vs medulla	Cortex: glomeruli + convoluted tubules. Medulla: loops of Henle + collecting ducts (pyramids).	**Filtration barrier**	Fenestrated endothelium + GBM (type IV collagen, negative charge) + podocytes.
Urinalysis normal	pH 5–7, SG 1015–1025, no protein, RBC < 5/HPF, no casts.	**Blood tests (KFTs)**	Urea & creatinine, electrolytes, albumin, cholesterol, ABG, CBC, C3.
Imaging	US first-line; X-ray for stones; VCUG for reflux; CT/MRI; DMSA for scars.	**Renal biopsy 'LIE'**	Light microscopy, Immunofluorescence, Electron microscopy.
Proteinuria	Abnormal protein in urine; normal ≤ 4 mg/m²/hr; nephrotic > 40 mg/m²/hr or UP/C > 2.	**Hematuria**	> 5 RBCs/HPF; gross (visible) or microscopic; glomerular (cola, casts) vs non-glomerular (red, clots).
Nephrotic syndrome	Heavy proteinuria + hypoalbuminemia (<2.5 g/dL) + hyperlipidemia (>200 mg/dL) + generalized edema (LEAP).	**Nephrotic-range proteinuria**	> 40 mg/m²/hr or UP/C ratio > 2 (normal ≤ 4 mg/m²/hr, P/C < 0.2).
Idiopathic NS	90% of childhood NS; MCNS is ~85% of idiopathic cases.	**MCNS**	Minimal change NS: age 2–6 yrs, selective proteinuria, normal C3, steroid-responsive.

Secondary NS	Due to systemic disease: SLE, diabetes, HSP, infections (HBV, HCV), drugs.
Urine in NS	Dark, foamy/frothy, 3+/4+ protein, selective (albumin).
Complications 'I THE H'	Infection (pneumococcal peritonitis), Thrombosis, Hypovolemia, Electrolyte disturbance, Hyperlipidemia.
Steroid therapy	Prednisone 2 mg/kg/day × 4 weeks then taper; resistant → biopsy + cyclosporine/tacrolimus.
Remission	Urine protein nil/trace for 3 consecutive days.
Relapse	Recurrence of proteinuria ≥ 3+ for 3 days — a common complication.
Diet in NS	Salt restriction, normal protein, low fat; fluid restriction only if severe edema.
Nursing priority	Daily weight, abdominal girth, I&O, skin care of edema, infection prevention.
APSGN	Most common cause of gross hematuria in children; after group A β-hemolytic strep (M types 4, 12, 49).
Latent period	Throat infection → 1–2 weeks; skin (impetigo) → 3–6 weeks.
Peak age	5–12 years; more in boys (≈2:1).
HEAL	Hematuria (cola/tea), Edema (periorbital), Azotemia/oliguria, Low C3 + hypertension.
Heart failure signs '3T'	Tachycardia, Tachypnea, Tender liver.
Hypertensive encephalopathy	Headache, vision changes, vomiting, confusion, convulsions — emergency.
Complications 'HER'	Hypertension/encephalopathy, Electrolyte (hyperkalemia), Renal failure, + CHF.
Urine in AGN	Cola/smoky, RBC casts, mild–moderate proteinuria.
C3 in APSGN	Low, returns to normal within 8 weeks; ASO / anti-DNase B raised.
Fluid rule	Intake = urine output + insensible loss (400 mL/m²/day).
Treatment 'DRIP'	Diet & rest (salt/fluid restriction), Relieve HTN & edema (diuretics, antihypertensives), ICU if needed, Problems/complications.
First sign of recovery	Diuresis (increased urine output); microscopic hematuria may last months.
Nephritis vs nephrosis	Nephritis = inflammation, hematuria, HTN, low C3. Nephrosis = massive proteinuria, hypoalbuminemia, foamy urine.
UTI	Bacterial growth in the normally sterile urinary system.
Lower vs upper UTI	Cystitis: little/no fever. Pyelonephritis: high fever, loin pain, toxic child.
Organisms 'KEEPS'	Klebsiella, E. coli (~80%), Enterococcus, Proteus, Staph & Pseudomonas.
Route	Mostly ascending; hematogenous in neonates.
Predisposing factors	Obstruction (PUV), VUR, hurried micturition, uncircumcised male, constipation.

Neonate/infant signs	Fever or instability, failure to thrive, vomiting/diarrhea, jaundice, irritability.
Clean-catch culture	> 100,000 CFU of one organism = diagnostic.
Urine bag	Non-invasive but high contamination — negative result useful, positive needs confirmation.
Imaging	Renal US; VCUG for VUR; DMSA for scars.
Pre-renal	Low perfusion: dehydration, shock, hemorrhage, heart failure.
Post-renal	Obstruction: PUV, stones, tumours.
Fluid challenge	If hypovolemic: normal saline bolus, then reassess urine output.
Calcium gluconate	1 mL/kg 10% slow IV with ECG monitoring — protects the heart.
Dialysis 'A-B-C-U-R'	Anuria/fluid overload, Bleeding, Creatinine rapidly rising, Uremic encephalopathy, Refractory K/acidosis.

Cystitis signs	Dysuria, urgency, frequency, enuresis, suprapubic pain.
SPA / catheter culture	Any growth = diagnostic; > 1 organism suggests contamination.
Treatment	Cystitis: oral 5 days (TMP-SMX, nitrofurantoin, amoxicillin). Pyelonephritis/infant: IV then oral 10–14 days.
AKI	Sudden fall in renal function (hours–weeks) → can't keep fluid, electrolyte, acid-base balance.
Renal (intrinsic)	ATN, AGN/HUS, nephrotoxic drugs.
Labs	↑ urea & creatinine, hyperkalemia, metabolic acidosis, ↑ phosphate, ↓ calcium.
Hyperkalemia 'C BIG K DROP'	Calcium gluconate, Bicarbonate, Insulin + Glucose, Kayexalate, Diuretics, salbutamol.
Diuretic trial	Furosemide if still oliguric after volume correction.
Nursing	Strict I&O, daily weight, restrict K & fluids, BP monitoring, watch for seizures.

Answer Key

الإجابات

1. c — Massive proteinuria
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

2. c — Minimal change disease
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

3. d — Inflammation is a main feature of nephritis
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

4. b — Weak immune system (infection)
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

5. d — Kidney biopsy
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

6. b — Autoimmune diseases
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

7. b — Atorvastatin
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). الـ statins بتستخدم في الـ hyperlipidemia المستمرة. الـ (atorvastatin) بيعالج المرض نفسه مش الكوليسترول prednisone.

8. c — Hypertension
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

9. c — Low-protein, low-sodium diet
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). في محاضرة الدكتور: بيتقلل البروتين AGN بياخد بروتين طبيعي أو زيادة شوية، والـ NS الـ azotemia. الاختيار c الملح بيتقلل في الاتنين. هو الأقرب من بس لو في الاختيارات.

10. d — Reduce inflammation
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

N1. c — Production of bile
Kidney: water, electrolytes, acid-base, waste excretion, BP (renin), EPO, vitamin D activation.

N2. b — Nephron
~1 million nephrons per kidney: glomerulus + tubules.

N3. a, b, c — Fenestrated endothelium; Glomerular basement membrane (negatively charged); Podocytes with slit diaphragms
3-layer shield: endothelium (size), GBM (charge), podocytes.

N4. a — ≤ 4 mg/m²/hr (P/C ratio < 0.2)
Nephrotic range: > 40 mg/m²/hr or P/C > 2.

N5. b — Renal ultrasound
US: size, echogenicity, stones, hydronephrosis. VCUG for reflux.

N6. a, b, c, e — Cola / smoky urine; RBC casts; Dysmorphic RBCs; Associated proteinuria
Clots and bright red urine point to non-glomerular (lower tract) bleeding.

N7. a, b, c — Light microscopy; Immunofluorescence; Electron microscopy
'LIE' approach.

N8. b — Erythropoietin
Loss of EPO in chronic kidney disease causes anemia.

N9. a — Normal urinalysis
Normal: SG 1015–1025, pH 5–7, no protein, RBC < 5/HPF.

N10. b — Voiding cystourethrogram (VCUG)
VCUG shows reflux and posterior urethral valves.

N11. a, b, c, e — Coagulation profile and platelets; Blood pressure; Signed consent; Hemoglobin / blood grouping
Main risk is bleeding; afterwards keep supine, monitor vitals and urine for hematuria.

N12. b — Falling blood pressure with rising pulse
Signs of hemorrhage/shock.

N13. b — The lower urinary tract
Glomerular hematuria is cola/smoky with RBC casts and no clots.

N14. a, b, c, e — Beetroot ingestion; Rifampicin; Urate crystals in infants; Food dyes
Pseudo-hematuria: foods, drugs, urates. APSGN gives true hematuria.

N15. a — Serum albumin 1.8 g/dL with heavy proteinuria
NS = heavy proteinuria + hypoalbuminemia (<2.5 g/dL) + hyperlipidemia + edema. Low C3 and RBC casts point to nephritis.

N16. a, b, c, e — Hypoalbuminemia < 2.5 g/dL; Cholesterol > 200 mg/dL; Generalized pitting edema; Heavy (nephrotic-range) proteinuria
Mnemonic LEAP: Lipids up, Edema, Albumin down, Proteinuria. Gross hematuria is not typical.

N17. b — Urine protein becoming negative / trace for 3 days
Remission = proteinuria disappears; weight falls as edema resolves.

N18. b — Spontaneous bacterial peritonitis
Infection (pneumococcal peritonitis) is a major complication because of IgG loss and steroid therapy.

N19. a, b, c, e — Daily weight at the same time; Measure abdominal girth; Turn frequently and protect edematous skin; Keep away from children with infections
Salt is restricted; protein intake is normal.

N20. b — Thrombosis (DVT, renal vein)
Mnemonic 'I THE H': Infection, Thrombosis, Hypovolemia, Electrolyte, Hyperlipidemia.

N21. a, b, c, d — Onset below 1 year or above 10 years; Persistent hypertension; Low C3; Gross hematuria
Typical MCNS (2–6 yrs, selective proteinuria, normal C3) is treated without biopsy.

N22. b — Test urine for protein daily and avoid live vaccines
Home dipstick detects relapse early; live vaccines are avoided during high-dose steroids; never stop steroids abruptly.

N23. a — In the first 3 months of life
Congenital: first 3 months; infantile: 3–12 months; childhood: after 1 year.

N24. b — 4–6 weeks of daily prednisone
Resistant cases need biopsy and other immunosuppressives.

N25. a — ≥ 2 relapses within 6 months or ≥ 4 in 12 months
Steroid-dependent = relapse during taper or within 2 weeks of stopping.

N26. b — Decreased plasma oncotic pressure from hypoalbuminemia
Albumin loss → fluid shifts to interstitium; secondary Na retention.

N27. b — Periorbital in the morning
Periorbital in the morning, shifts to dependent parts during the day; may become generalized (anasarca).

N28. b — The liver increases lipoprotein synthesis in response to low albumin
Compensatory hepatic synthesis plus reduced lipoprotein lipase.

N29. a, b, c, d — Hypertension; Hyperglycemia; Cushingoid face and weight gain; Growth retardation
Also infection, cataract, osteoporosis, mood changes.

N30. b — Monitor for fluid overload: BP, respiration, crackles
Albumin draws fluid into vessels; risk of pulmonary edema/hypertension.

N31. a — Pneumococcal vaccine
Pneumococcus is the main cause of peritonitis in NS.

N32. a — Excess fluid volume
Physiological needs first; also risk for skin breakdown and infection.

N33. a — Sore throat 1–2 weeks ago
Throat infection precedes APSGN by 1–2 weeks; skin infection (impetigo) by 3–6 weeks.

N34. a, b, c, e — Cola / tea-colored urine; Periorbital edema; Hypertension; Oliguria
HEAL: Hematuria, Edema, Azotemia/oliguria, Lab low C3 + hypertension. Proteinuria is mild–moderate.

N35. b — Notify the physician — suspect hypertensive encephalopathy
Hypertensive encephalopathy (headache, vision changes, convulsions) is an emergency complication.

N36. a — Low C3 with raised ASO titre
C3 is low and returns to normal within ~8 weeks; ASO/anti-DNase B show recent strep.

N37. b — Previous day's urine output + insensible loss (≈400 mL/m²/day)
Intake = output + insensible loss, with salt restriction (DRIP: Diet & Rest).

N38. b — Increased urine output (diuresis)
Diuresis comes first; microscopic hematuria may persist for months.

N39. a, b, c, d — Daily weight and strict I&O; BP monitoring every 4 hours; Low-salt diet; Bed rest during the acute oliguric phase
Potassium is restricted when oliguric (risk of hyperkalemia).

N40. a — C3 still low after 8 weeks or normal at onset
Biopsy if: rapid progression, normal C3, hematuria > 6 months, persistent low C3.

N41. b — Immune complex deposition in the glomeruli
Antigen–antibody complexes deposit → complement activation → inflammation (type III hypersensitivity).

N42. b — Salt and water retention from decreased GFR
Reduced GFR → fluid overload → hypertension, edema, circulatory congestion.

N43. a, b, c — Hypertensive encephalopathy; Acute heart failure; Acute kidney injury with hyperkalemia
DRIP: ICU if renal failure, heart failure or encephalopathy.

N44. b — Eradicate remaining streptococci and prevent spread
Penicillin (or erythromycin) does not change the renal course.

N45. a — Most children (> 95%) recover completely
Gross hematuria clears in 1–3 weeks; microscopic hematuria can last up to 1–2 years.

N46. b — Bananas and oranges
High-potassium foods are restricted during oliguria.

N47. a, b, c — Gross hematuria; Hypertension; Low C3
Massive proteinuria and hyperlipidemia belong to nephrotic syndrome.

N48. a — Daily weight on the same scale
Combined with strict intake and output.

N49. b — Escherichia coli
E. coli causes ~80% of UTIs ('KEEPS' bugs).

N50. b — Fever or temperature instability, poor feeding, jaundice
Infants show non-specific signs: fever, failure to thrive, vomiting, jaundice.

N51. a, b, c, d — Vesicoureteral reflux; Posterior urethral valves; Uncircumcised male infant; Constipation
Obstruction, VUR, hurried micturition, uncircumcised males, constipation.

N52. b — Any bacterial growth
Any growth on SPA/catheter = UTI; clean-catch needs > 100,000 CFU; > 1 organism = contamination.

N53. b — Void regularly and completely; drink enough fluids
PREVENT UTI: hydration, regular complete voiding, treat constipation, wipe front to back.

N54. a, b, d — High fever with rigors; Loin pain / tenderness; Vomiting and toxic look
Cystitis: dysuria, urgency, frequency, little or no fever.

N55. c — Urine bag
Bag specimens are useful mainly when negative.

N56. a — Recurrent pyelonephritis and renal scarring
Scarring can lead to hypertension and chronic kidney disease.

N57. a, b, c, e — Age below 3 months; Toxic appearance or vomiting; Suspected pyelonephritis with dehydration; Urosepsis
Older children with cystitis can be treated orally.

N58. b — Clean the genital area, discard the first part, then collect midstream
Send to the lab within 1 hour or refrigerate.

N59. a — Positive leukocyte esterase and nitrites
Pyuria (> 5 WBC/HPF) and bacteriuria; culture confirms.

N60. a — Low-dose nitrofurantoin or TMP-SMX at night
Given once at bedtime; plus treating constipation and regular voiding.

N61. a — Pre-renal
Pre-renal = decreased perfusion (dehydration, shock, hemorrhage).

N62. b — Calcium gluconate 10% slowly IV
Calcium gluconate stabilizes the myocardium first; then shift K (insulin+glucose, bicarbonate, salbutamol) and remove it.

N63. a, b, c, d — Fluid overload with pulmonary edema; Refractory hyperkalemia; Uremic encephalopathy; Severe metabolic acidosis not responding
A-B-C-U-R plus refractory acidosis and hyperkalemia.

N64. b — Strict intake and output with daily weight
Fluid balance guides therapy; restrict K and fluids.

N65. a, b, c, e — Raised urea and creatinine; Metabolic acidosis; Hyperkalemia; Hypocalcemia
Phosphate rises (hyperphosphatemia) with low calcium.

N66. c — Post-renal (obstructive)
Obstruction distal to the kidney = post-renal; ultrasound first.

N67. a — 0.5–1 mL/kg/hr
< 1 mL/kg/hr in infants, < 0.5 mL/kg/hr in children.

N68. a, b, c, e — Aminoglycosides (gentamicin); NSAIDs; Contrast media; Amphotericin B
Monitor drug levels and creatinine; ensure hydration.

N69. a — Hemolytic anemia, thrombocytopenia and renal failure after bloody diarrhea
Usually E. coli O157:H7 — an intrinsic renal cause.

N70. a — Peaked (tall tented) T waves
Later: widened QRS, sine wave, arrest.

N71. b — Dehydration and hypokalemia
Large urine volumes cause water and electrolyte loss.

N72. a — It is technically simple and needs no vascular access
Watch for peritonitis: cloudy effluent, fever, abdominal pain.

N73. a, b, c, e — Adequate calories mainly from carbohydrate and fat; Restricted potassium; Restricted phosphate; Controlled protein
Enough calories prevent catabolism, which raises urea and K.